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GUIA 3 parcial CIRCUITOS LOGICOS IF324

P1. Obtenga la diferencia de los siguientes Binarios 10101111 y 11010011

$$\begin{array}{r} 10101111_2 - 11010011_2 \\ 175 - 171 = 36_{10} = 100100_2 \\ \hline 36 \end{array}$$

P2. Obtenga la diferencia Hexadecimal de F3B51 y ACD19

$$\begin{array}{r} F3B51_{16} - ACD19_{16} \rightarrow 998225 \\ \hline 707865 \\ \hline 290360 \end{array} \left. \vphantom{\begin{array}{r} F3B51_{16} - ACD19_{16} \end{array}} \right\} \text{Decimal}$$

$$\begin{array}{r} F3B51 \\ - ACD19 \\ \hline 46E38_{16} \end{array}$$

Hexadecimal

P3. Obtenga la diferencia en BCD 0111 0100 0110 0110 y 1001 1000 0111 0101

$$\begin{array}{r} 7466 \\ - 9875 \\ \hline -2409_{10} \end{array} \rightarrow \begin{array}{cccccccc} \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ 7 & 4 & 6 & 6 & 9 & 8 & 7 & 5 \end{array}$$

$$= 0010 \ 0100 \ 0000 \ 1001_{BCD}$$

$$= -2409_{BCD}$$

P4. Obtenga la conversión (110101010)₂ a Gray

$$\begin{aligned} g_1 &= b_1 = 1 \\ g_2 &= b_1 \oplus b_2 = 1 \oplus 1 = 0 \\ g_3 &= b_2 \oplus b_3 = 1 \oplus 0 = 1 \\ g_4 &= b_3 \oplus b_4 = 0 \oplus 1 = 1 \\ g_5 &= b_4 \oplus b_5 = 1 \oplus 0 = 1 \\ g_6 &= b_5 \oplus b_6 = 0 \oplus 1 = 1 \\ g_7 &= b_6 \oplus b_7 = 1 \oplus 0 = 1 \end{aligned}$$

$$g_8 = b_7 \oplus b_8 = 0 \oplus 1 = 1$$

$$g_9 = b_8 \oplus b_9 = 1 \oplus 0 = 1$$

$$\rightarrow 101111111 \text{ Gray}$$

P5. Obtenga la conversión $(110101010)_{\text{Gray}}$ a (binario)

$$\begin{aligned} b_1 &= g_1 = 1 \\ b_2 &= b_1 \oplus g_2 = 1 \oplus 1 = 0 \\ b_3 &= b_2 \oplus g_3 = 0 \oplus 0 = 0 \\ b_4 &= b_3 \oplus g_4 = 0 \oplus 1 = 1 \\ b_5 &= b_4 \oplus g_5 = 1 \oplus 0 = 1 \\ b_6 &= b_5 \oplus g_6 = 1 \oplus 1 = 0 \\ b_7 &= b_6 \oplus g_7 = 0 \oplus 0 = 0 \end{aligned}$$

$$b_8 = b_7 \oplus g_8 = 0 \oplus 1 = 1$$

$$b_9 = b_8 \oplus g_9 = 1 \oplus 0 = 1$$

$$b = 100110011_2$$

P6. Obtenga el cociente de 10111111 y 1011

$$\begin{array}{r} 10001 \\ 1011 \overline{) 10111111} \\ \underline{1011} \\ 000011 \\ 1 \\ 11 \\ 111 \\ 1111 \\ \underline{1011} \end{array}$$

P7. Transmítala palabras **Muchos éxitos** (Impar)

M 1001101 U 1010101 C 1000011 H 1001000 O 1001111 S 1010011

E 1000000 X 1100101 I 1011000 T 1001001 O 1010100 I 1010101

P8. Transmítala palabras **FELIZ NATIVIDAD** (par)

F 1000110 E 1000101 L 1001100 I 1001001 Z 1011010

N 0100000 A 1001110 T 1000001 I 1010100 V 10011001 I 1010101

D 1001001 A 1000001 D 1001001

P9. Multiplique los siguientes binarios: 111011101×10101

$$\begin{array}{r}
 111011101 \times (1 \times 10101101) \\
 000000000 \times (0 \times 11101101) \\
 11101110100 \times (1 \times 11101110) \\
 000000000000 \times (0 \times 11101110) \\
 1110111010000 \times (1 \times 11101110)
 \end{array}$$

$$\begin{array}{r}
 111011101 \\
 000000000 \\
 11101110100 \\
 + 000000000000 \\
 \hline
 111011101000 \\
 \hline
 100001100010101
 \end{array}$$

P10. Divida 101111101 entre 11101

$$\begin{array}{r}
 10124 \\
 11101 \overline{) 101111101} \\
 \underline{11101} \\
 0101110 \\
 \underline{0101101} \\
 00010002
 \end{array}$$

P11. 11100110110_2 (Usando método 5, +3) CONVERTIR A CODIGO BCD

$$\begin{array}{l}
 1 \rightarrow 0001 \\
 8 \rightarrow 1000 \\
 4 \rightarrow 0100 \\
 6 \rightarrow 00110
 \end{array}
 \quad R//: 0001 \ 1000 \ 0100 \ 0110_{BCD}$$

P12. Obtenga la conversión $(653415)_7$ a (decimal)

$$\begin{aligned}
 &= 6 \cdot 7^5 + 5 \cdot 7^4 + 3 \cdot 7^3 + 4 \cdot 7^2 + 1 \cdot 7^1 + 5 \cdot 7^0 \\
 &= 6(16807) + 5(2401) + 3(343) + 4(49) + 1(7) + 5(1) \\
 &= 100842 + 12005 + 1029 + 196 + 7 + 5 \\
 &= 114084_{10}
 \end{aligned}$$